Written Homework Assignment #2

Problems from Ch 5 and 6

You should have read the selections from chapters 5 and 6 of the Stephens text, as specified on the readings page. Having done so will assist you in completing the following assignment.

Note that some of the problems in these assignments are from chapters of the book which you have not been 'officially' assigned to read in detail. However, simply skimming through those chapters ***will*** allow you to find the information which will help you answer the problems. This will serve TWO purposes: 1) you will get a bit of research experience as you scan through the chapters; and 2) you'll probably remember a bit more information out of them because you are looking for something specific.

Here are the problems for easy reference.

Problem 5.1, Stephens page 116

What's the difference between a component-based architecture and a service-oriented architecture?

Component-based architecture treats parts of the system as components that provide services for each other. Service-oriented architecture is similar but parts run on separate computers communicating over a network. Parts are more separated in service-oriented architecture.

Problem 5.2, Stephens page 116

Suppose you're building a phone application that lets you play tic-tac-toe against a simple computer opponent. It will display high scores stored on the phone, not in an external database. Which architectures would be most appropriate and why?

All you would need is monolithic architecture, as it is a simple, relatively small and contained application.

Problem 5.4, Stephens page 116

Repeat question 3 [after thinking about it; it repeats question 2 for a chess game] assuming the chess program lets two users play against each other over an Internet connection.

The main changes are that the program must communicate with another instance across the Internet. This would make it a monolithic service-oriented application.

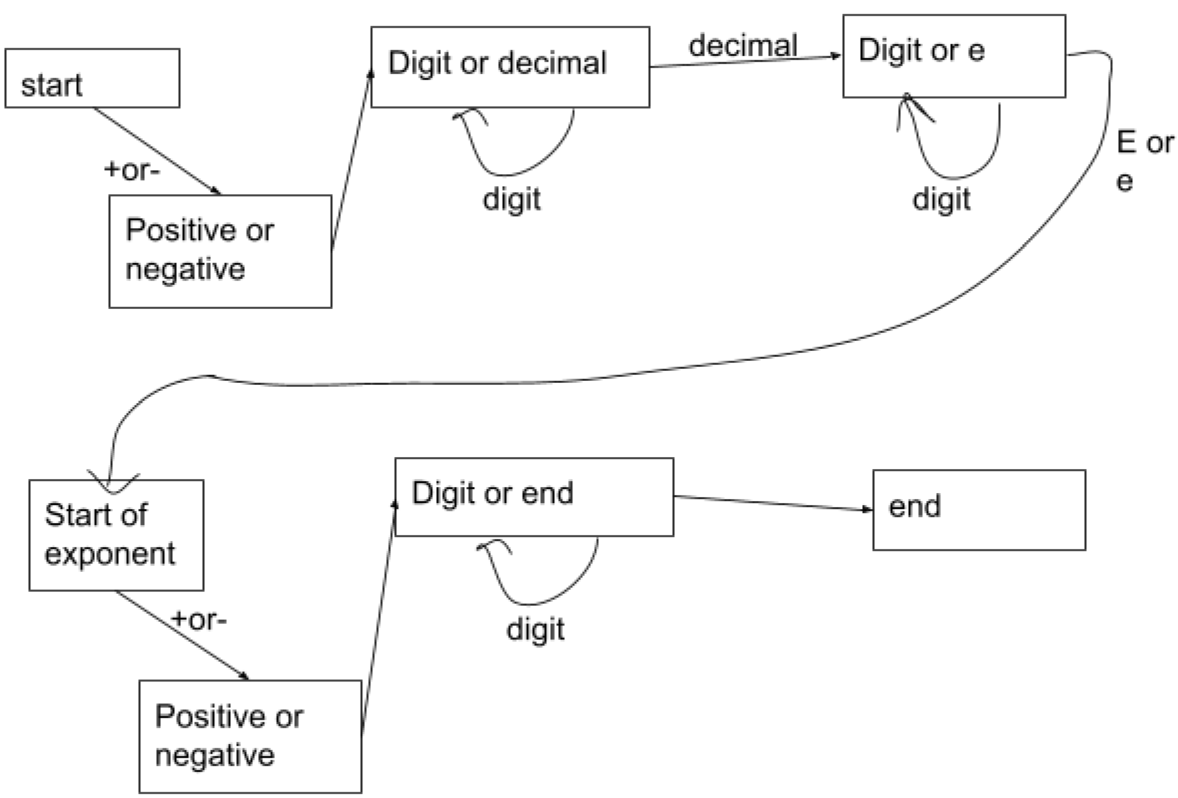
Problem 5.6, Stephens page 116

What kind of database structure and maintenance should the `ClassyDraw` application use?

ClassyDraw should use a separate file to store each drawing so it doesn't require much of a database. Every time a file is edited a temporary file is made so it can be recovered in case of a crash.

Problem 5.8, Stephens page 116

Draw a state machine diagram to let a program read floating point numbers in scientific notation as in +37 or -12.3e+17 (which means -12.3 x 10^{17}). Allow both E and e for the exponent symbol. [Jeez, is this like Dr. Dorin's DFAs, or *what*???]



Problem 6.1, Stephens page 138

Consider the `ClassyDraw` classes `Line`, `Rectangle`, `Ellipse`, `Star`, and `Text`.

1. What properties do these classes all share?

   These classes share properties related to drawing, such as foreground color, background color, width, height.

2. What properties do they **NOT** share?

They do not share properties specific to the class such as number of points for Star, font for Text.

3. Are there any properties shared by some classes and not others?

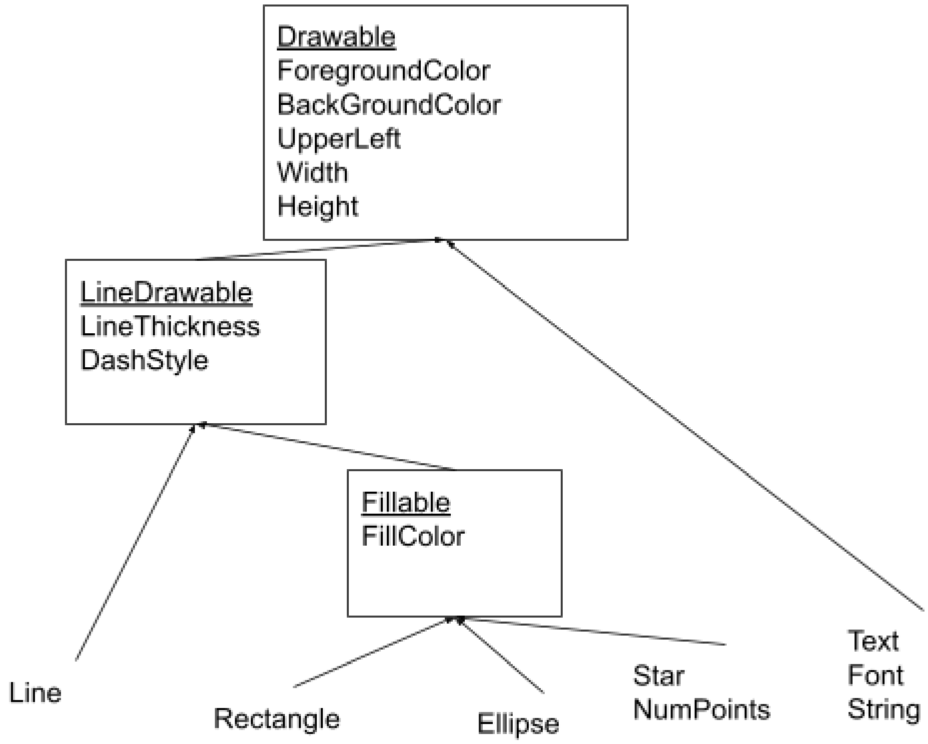
Yes, fill color for Rectangle, Ellipse, Star, and line thickness, dash style for Line, Rectangle, Ellipse, Star.

4. Where should the shared and nonshared properties be implemented?

Shared properties should go in a higher class such as Shape, nonshared should go in the individual class.

Problem 6.2, Stephens page 138

Draw an inheritance diagram showing the properties you identified for Exercise 6.1. [Create parent classes as needed, and don't forget the `Drawable` class at the top.]


Drawable
ForegroundColor
BackGroundColor
UpperLeft
Width
Height
LineDrawable
LineThickness
DashStyle
Fillable
FillColor
Line
Rectangle
Ellipse
Star
NumPoints
Text
Font
String